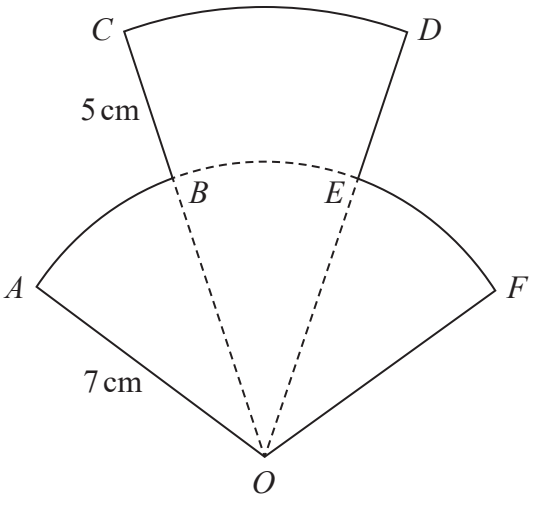
19 Here is a logo, *OABCDEF*.



OABEF is a sector of a circle, centre *O* and radius 7 cm.
OBCDE is a sector of a circle, centre *O* and radius 12 cm.

Angle *AOF* = 100°
Angle *COD* = 40°

The perimeter of the logo is *P* cm.

Find the exact value of *P*.
Give your answer in the form $a + b\pi$ where *a* and *b* are integers.

Arc CD = $\frac{40}{360} \times 2\pi \times 12 = \frac{8\pi}{3}$

1 mark
Outer arc, radius 12.

Arcs AB + EF = $\frac{100 - 40}{360} \times 2\pi \times 7$

1 mark
Only exposed inner arcs.

$= \frac{7\pi}{3}$

Straight edges = 7 + 7 + 5 + 5 = 24

1 mark
Add the outline only.

Perimeter = $24 + \frac{8\pi + 7\pi}{3} = 24 + 5\pi$

1 mark
Exact length in cm.

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(Total for Question 19 is 4 marks)

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